

SGD

Spatial Gesture Design — binaural instrument for VST3, AU and Standalone.

USER MANUAL · VERSION 0.1.0 · RADICALPROCESS

SGD turns a sound file into a *gesture*: a sound that travels a path through a virtual room around the listener. You draw the path, decide how fast the sound moves along it, mix it, and bind it to a MIDI key. Pressing that key plays the sound and moves it — rendered binaurally, for headphones.

1. Concepts

Three ideas carry the whole plugin. Everything in the interface is one of them, or a view of one.

1.1 Rooms, gestures, keys

A **room** is the space the listener stands in the middle of. It has a name, a size in metres, and wall acoustics. A room also owns everything inside it: its gestures and their sound files. Only one room is open at a time, and rooms are independent — switching rooms swaps out the entire contents of the plugin.

A **gesture** is a sound plus its motion. It holds:

- a **sound file**, optionally trimmed to a shorter excerpt with fades;
- a **trajectory** — a curved path through the room, drawn from above, with a height for every point on it;
- a **motion curve** — how playback time maps onto position along the path, so the source can accelerate, hesitate, stop or run backwards;
- a **stereo width curve** (stereo sounds only) — how far the left and right channels spread apart over time;
- a **mixer strip** — level, three-band EQ and reverb send;
- a **key**, a name and a colour.

A **key** is a MIDI note, 0–127. Each gesture sits on exactly one key, and no two gestures in a room may share one, so a room holds at most 128 gestures. The key is how a gesture is played, whether the note comes from the on-screen keyboard or from your DAW.

1.2 Why rendering happens

Spatialising a moving source properly is expensive, so SGD does it **ahead of time**. Whenever a gesture's sound, path, motion, width or render options change, that gesture is queued for an offline binaural render in a background thread. Playback then streams the finished stereo file, which costs almost nothing.

Two consequences follow, and they explain most of what you will see while working:

- A gesture is not *immediately* audible in its spatialised form after an edit. It shows QUEUED, then a spinner while rendering, then READY. Watch the Render panel for progress.
- The reverb tail is *not* pre-rendered. Each gesture's dry signal is sent to one shared room reverb through its Send knob, and that reverb follows the room's size and wall material live.

HEADPHONES

SGD's output is binaural — it renders what your two ears would hear. It is designed to be monitored on headphones. Over loudspeakers, the spatial impression largely collapses.

2. Installation & setup

2.1 Formats and hosts

SGD builds as three products from the same code:

FORMAT	PLATFORM	NOTES
VST3	macOS, Windows	Instrument (synth), MIDI in, stereo out.
AU	macOS	Instrument. Component installed to the usual AU folder.
Standalone	macOS, Windows	Own application window; audio and MIDI devices are chosen in its own settings.

SGD is an **instrument**, not an effect. It never passes host input audio to its output, even where the host provides an input bus. Load it on an instrument track and send MIDI to it.

The output is stereo. Mono and stereo output layouts are accepted; input and output layouts must match, which is what hosts request for an instrument.

LATENCY

If your session does not run at 48 kHz, SGD converts its final mix from 48 kHz to the host rate and reports the conversion latency to the host, so delay compensation keeps SGD in time with the rest of the session. At 48 kHz the conversion is bypassed and latency is zero.

2.2 First launch

SGD cannot do anything without a room, so the very first time you open the editor SGD saves one for you: a standard 12 × 12 × 12 m room named **My First Room**. Nothing is asked of you — add a gesture and start drawing straight away. Rename it, resize it or change its walls whenever you like, from the room editor.

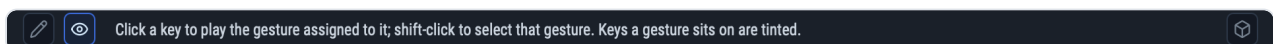
From then on, opening the plugin re-opens the room your session refers to — or the first saved room if the session does not name one. The one case where SGD does insist on a room is when you delete the last one you have: the **room editor overlay then opens automatically and cannot be cancelled** until you give it a name, a size, and press **Save**.

3. Interface tour



The edit view. Three columns — gestures, editor, inspector — over the keyboard. The room here is *Demo Hall* (14 × 7 × 18 m) with five gestures; *flyby* is selected, and its path is drawn from above in the middle column with the waveform along it.

3.1 Header strip



The header, with the mouse over the keyboard: the status line has replaced the view name with that control's description. Hover anything and its own sentence appears here.

The strip across the top holds, from left to right:

CONTROL	WHAT IT DOES
Edit button	Shows the edit view : build, mix and render the gestures of this room.
Eye button	Shows the observe view : watch the gestures move through the room in 3D.
Status line	Reads out the tooltip of whatever the mouse is over (see below).
Room button	Opens the room menu: new, open, edit, delete.

3.2 Body

Below the header, exactly one body view is visible at a time:

- **Edit view** — three columns at 30 % / 40 % / 30 % of the width: the gesture panel, the editor for the selected gesture, and the inspector. If no gesture is selected, the middle and right columns are empty.
- **Observe view** — a single 3D scene of the room; see Observe view.

Two overlays can spread across the whole window and dim what is behind them: the room editor and the trim editor. While an overlay is open, nothing behind it can be clicked.

3.3 Keyboard & range strip



The range strip and keyboard. Ticks on the slot map mark the five assigned keys, the lit band is the window on show, and each assigned key carries its gesture's colour. A4 is washed blue because flyby is the selected gesture.

Along the bottom sits a piano keyboard showing four octaves at a time, with a thin **range strip** above it:

- **Arrows** at each end move the keys on show one octave down or up.
- The **slot map** in the middle represents all 128 keys. A tick marks a key a gesture sits on; the lit band shows which keys are currently on screen. Drag the band to move the window.
- The **readout** on the right names the range on show, e.g. C3–B6 · 128 slots.

3.4 Reading the status line

SGD does not use pop-up tooltips. Every control has a one-sentence description, and it is printed in the **status line in the header** when you hover the control. This keeps text from covering the 3D

views, and gives long explanations room to be read. If you are unsure what something does, hover it and read the top of the window.

TIP

The status line is the fastest reference in the plugin. Almost every sentence in the tables of this manual is the same sentence you will find there.

4. Quick start

From a blank slate to a moving sound in eight steps.

- 1 Take the room you were given.** On first launch *My First Room* is already there, so skip straight to the next step. For a room of your own, press the room button and choose *New Room*. Name it, pick a size preset (or set Width / Height / Depth by hand), pick a wall material, and press **Save**.
- 2 Add a gesture.** In the gesture panel, press **+**. A card appears with a free key already assigned and a colour that contrasts with its neighbours.
- 3 Give it a sound.** Press **Load** on the card and pick a mono or stereo .wav / .aiff file — or just drag the file onto the card.
- 4 Select the gesture** by clicking its card. The editor and inspector columns fill in.
- 5 Draw the path.** In the top view, click empty space to append a node. Repeat to build a path. Drag a node to move it, drag its handles to curve the segment.
- 6 Shape it in 3D.** Open the *Elevation* tab and drag nodes up or down to lift parts of the path off the floor.
- 7 Wait for the render.** The card shows QUEUED, then a spinner, then READY. The Render panel shows the progress.
- 8 Play it.** Click the tinted key on the on-screen keyboard, or send that MIDI note from your DAW. Listen on headphones.

From there: refine timing in the *Motion* tab, balance the gesture with the *Mix* knobs, and dial in the room's reverb tail with the Send knob and the room's wall material.

5. Rooms

5.1 Room menu

The room button in the header opens a menu with four entries:

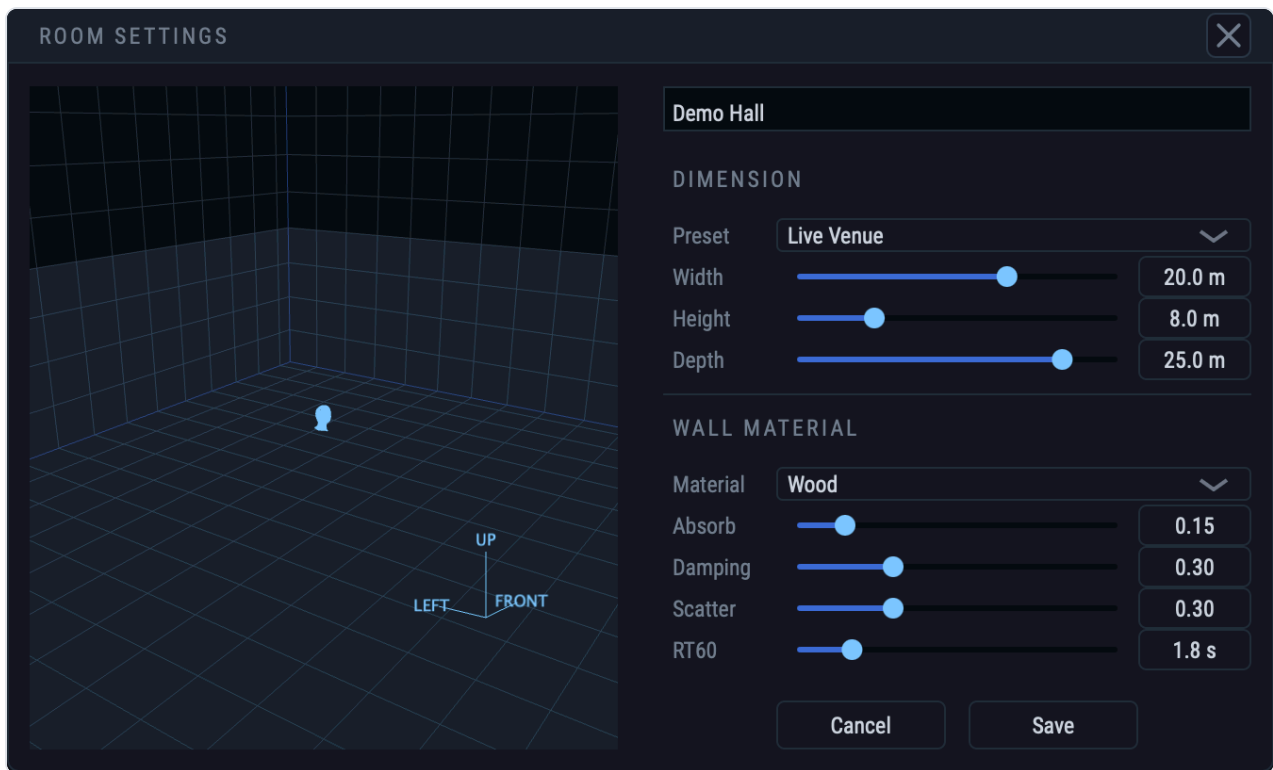
ENTRY	EFFECT
New Room	Opens the room editor empty, to create and save a new room.
Open Room	A submenu of every saved room. Choosing one closes the current room and loads it, with all its gestures and sounds.
Edit Room	Opens the room editor on the current room, so its name, size and acoustics can be changed.
Delete Room	Deletes a saved room and everything in it. You are asked to confirm first.

CAREFUL

Deleting a room deletes its folder on disk: its `room.json`, its imported sound files, and every rendered file. This cannot be undone from inside SGD.

5.2 Room editor

The room editor is an overlay that dims the rest of the window. On its left is a live 3D preview of the room seen from where the listener stands — **drag to orbit, scroll to zoom**. On its right, from top to bottom: the room name, the *Dimension* group, the *Wall material* group, and the Cancel / Save buttons.



The room editor. The preview on the left redraws as the sliders move — here the *Live venue* size preset (20 × 8 × 25 m) with *Wood* walls. Change any slider by hand and the dropdown above it falls back to *Custom*.

- **Save** stores the room under this name and size and opens it. It stays unavailable until the room has a name.
- **Cancel** (and the × in the corner) closes without saving. It is unavailable while no room exists at all — which is why the room demanded after you delete your last one is mandatory.

5.3 Dimension

Three sliders set the room's size in metres, each from 1 m to 30 m: **Width** (left to right), **Height** (floor to ceiling) and **Depth** (front to back). The listener always stands at the centre.

The **Preset** dropdown sets all three at once to a typical space — see the table in the reference. Moving any slider by hand deselects the preset, which then reads as *Custom*.

The room's size matters in three ways: it is the box the trajectory is drawn inside (a normalised path is stretched onto the real metres), it feeds the reverb, and it is what the Stats panel measures distances and speeds against.

5.4 Wall material

Four sliders describe the acoustics of the walls. The **Material** dropdown sets all four to a rough reference for a surface; tweaking any of them by hand deselects it.

CONTROL	RANGE	DEFAULT	MEANING
Absorb	0 – 1	0.20	How much sound the walls swallow per bounce. 0 is mirror-hard; 1 swallows everything.
Damping	0 – 1	0.40	How much faster the highs die away than the lows — soft rooms sound darker as they ring out.
Scatter	0 – 1	0.50	How rough the walls are. Rough surfaces diffuse the reverb; smooth ones keep it even.
RT60	0.1 – 10 s	1.5 s	How long the tail rings: the time it takes to fall away by 60 dB.

LIVE

The room's dimensions and acoustics feed the shared reverb *immediately* — no re-render is needed to hear a change to Absorb, Damping, Scatter or RT60. The early reflections baked into each gesture's binaural render are a different matter: those come from the Render options and do require a re-render.

6. Gestures

6.1 The gesture panel

GESTURES

5 / 128

Search

A4



flyby ×3

drumLoop.wav [s] · 4.8s



C5



orbit ×3

drumLoop.wav [s] · 4.8s



E5



rise ×2

voice.wav [m] · 2.5s



G5



swoop ×2

voice.wav [m] · 2.5s



B5



drift ×3

drumLoop.wav [s] · 4.8s



The gesture panel. Five gestures, the slot counter reading 5 / 128, and the add / duplicate / delete buttons along the bottom.

The left column lists every gesture in the room, one card per gesture. At the top:

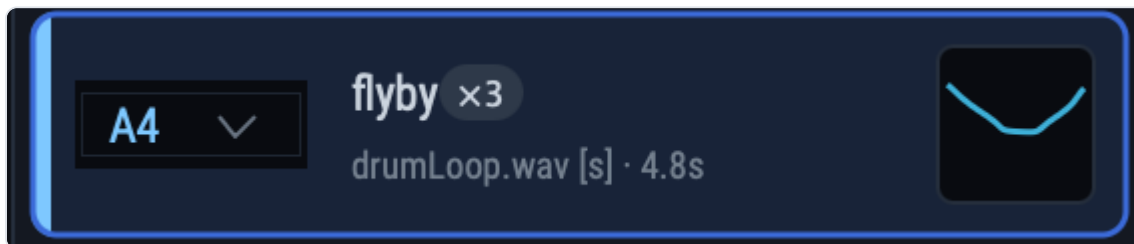
- a **search field** — type to show only gestures whose name contains what you typed;
- a **slot counter** — how many of the 128 key slots this room's gestures take.

At the bottom, three buttons:

BUTTON	EFFECT
+	Adds an empty gesture to the room, on the first free key.
Duplicate	Copies the selected gesture — sound and path included — onto a free key.
Delete	Deletes the selected gesture.

Clicking a card selects that gesture; the editor and inspector columns then show it.

6.2 Anatomy of a card



One card. Left to right: colour stripe, key chip (A4), name with its ×3 usage badge, the sound file line, and the path thumbnail. The blue outline marks it as the selected gesture.

ELEMENT	WHAT IT IS
Colour stripe	Down the left edge. Click it to open a palette and recolour the gesture. The colour identifies the gesture in the trajectory views and the observe view, and tints its key on the keyboard.
Key chip	The key that triggers this gesture, as a note name. Open it to reassign; keys taken by another gesture are greyed out.
Name	Click to rename, up to 20 characters.
Subtitle	The sound file, whether it is mono [m] or stereo [s], and the length of the trimmed sound, e.g. bell.wav [s] · 2.4s.
Load / drag hint	Shown instead of the subtitle while the gesture has no sound.
Usage badge	A small ×2 pill next to the name, shown when more than one gesture shares the same imported sound file.
Status badge	QUEUED, a turning spinner while rendering, or READY. No badge means idle — nothing pending.
Thumbnail	The path this gesture travels, seen from above.

6.3 Giving a gesture a sound

Two ways, and they behave identically:

- press **Load** on the card and pick a file;
- drag one file straight onto the card. The card lights up when the file is acceptable.

SGD accepts **one mono or stereo .wav, .aiff or .aif file**. Anything else — more than two channels, another format, several files at once — is refused, with a message saying why. Common source rates (44.1, 48 and 96 kHz) are handled; see Sample rates.

On import the file is copied into the room's own folder, so the gesture no longer depends on where the original lived. If two gestures import the same file, it is stored once and reference-counted — hence the ×2 usage badge.

6.4 Keys and slots

Keys run over the full MIDI range, C-1 (note 0) to G9 (note 127) — 128 slots. New gestures are assigned a free key starting from A4 (note 69) and working outwards, so the first gestures land under the middle of a keyboard.

No two gestures may share a key. When you open a card's key chip, the keys already taken are greyed out, so a clash cannot be created by accident.

7. The gesture editor

The middle column edits the selected gesture's motion through the room. It is one large **top view**, a tab area with three pages below it, and a transport strip at the bottom.



The editor column with the elevation tab open. The top view carries the numbered trajectory nodes and the waveform drawn along the path; the room's footprint is labelled (14.0 m across, 18.0 m deep). Dotted guides link each node to its counterpart in the elevation below.

7.1 Top view (trajectory)

The path this gesture travels, seen from above, with the listener at the centre. The frame follows the room's footprint, so a wide, shallow room draws as a wide, shallow rectangle.

ACTION	RESULT
Click empty space	Appends a node at that point, extending the path.
Drag a node	Moves it.
Drag a node's handle	Curves the segments meeting at that node.
Shift-click a node	Removes it. A gesture always keeps at least one node.
Clear button	Removes every node and starts the path again.
Waveform button	Shows or hides the sound's waveform drawn along the path — the loudness of the sound at each point in space.

During playback a marker walks the path in the gesture's colour; for a stereo gesture, two markers labelled L and R separate according to the stereo width curve. Editing is disabled while the gesture is playing.

7.2 Elevation tab

The same path seen from the side: how high above the floor each node sits. The horizontal axis lines up with the top view above it, and faint dotted guides connect each node to its counterpart there.

- Drag a node **up or down** to raise or lower the path at that point.
- Drag it **sideways** to slide it across the room.
- Shift-click a node to remove it.
- Nodes are *added* in the top view only.

The vertical range is the room's height, so a taller room gives the same drag more metres of travel. Between nodes the height rises in a straight line, while the ground-plane path curves.

7.3 Motion tab

The motion curve maps **playback time** (left to right) onto **position along the path** (bottom to top). A straight diagonal from corner to corner — the default — sweeps the path once at constant

speed.



The motion tab for fLyby. The shallow ends and steep middle mean the source eases in, races past the listener, then eases out. The waveform behind is the sound re-binned to position, and the numbered horizontal lines mark where the five trajectory nodes fall along the path.

ACTION	RESULT
Click	Adds a node.
Drag	Moves a node. The endpoints stay pinned at the corners.
Shift -click	Removes an interior node.

The **slope** of the curve is the speed:

- steep — the source races along the path;
- flat — it stands still while the sound continues;
- downward — it runs backwards along the path;
- a step up — it jumps.

The tab also draws the sound's waveform re-binned to position, and reference lines marking where the trajectory's nodes fall along the path, so you can time an arrival at a specific corner.

DOPPLER

Speed here has an audible consequence: a fast-moving source bends in pitch. If the pitch sweep is too strong, either flatten the motion curve or turn down the **Doppler** knob in the Render panel.

7.4 Stereo width tab

Available for stereo gestures only; the tab button is disabled for a mono sound. The curve says how far the two channels stand apart over playback time: each channel is the centre source rotated around the listener by half the angle, to its own side.



The stereo width tab for flyby. The two traces are the left and right sources, drawn as mirror images either side of the centre line; they open out to their widest half-way through and close again by the end. The axis runs from 0s to the sound's length, 4.8s.

- 0° — both channels ride the path together, exactly as a mono gesture would.
- Up to 180° — the widest spread, each source a quarter turn from the centre.
- Dragging a node **across the centre line** makes the angle negative, which *flips* left and right.

Editing works as in the motion tab: click to add, drag to move, **Shift**-click to remove.

7.5 Transport

The strip at the bottom of the editor previews the selected gesture on its own:

ELEMENT	WHAT IT DOES
Name	The gesture this transport plays.
Play / pause	Plays or pauses. Enabled once the gesture has a sound.
Stop	Stops playback and rewinds to the start.
Time	Playback position over the length of the trimmed sound.

8. The inspector

The right column stacks five panels for the selected gesture and the room's output.

RENDER

☒ Direct

☒ 1st

☒ 2nd

☐ 3rd

100

Doppler

0%

STATS

Path length 18.2 m

Ascent / descent 0.9 m / 0.5 m

Avg / peak speed 3.8 / 8.7 m/s

Accel / decel 1.8x / 1.8x

Spatial centroid 0.1 m right, 2.6 m front

SOURCE



drumLoop.wav



MIX

-2

0

0

0

-14

Gain

Low

Mid

High

Send

MASTER

0 dB

L

R

The inspector: Render, Stats, Source, Mix, Master. The Stats figures are the real measurements of the flyby path in this 14 × 7 × 18 m room.

8.1 Render

Gestures are rendered to binaural audio in the background; this panel says what the render is made of and how far it has got.

What is mixed in

TOGGLE	DEFAULT	MIXES IN
Direct	on	The sound travelling straight from the source to the listener.
1st	on	Reflections that bounced off one wall.
2nd	on	Reflections that bounced off two walls.
3rd	off	Reflections that bounced off three walls.

Higher orders add realism and render time. Turning *Direct* off leaves only reflections, which is a way to push a source behind a wall or make it read as distant.

Doppler

A dial from 0 % to 100 % (default 100 %): how much of the pitch shift a moving source makes the render keeps. At 100 % the render is unrestricted; lower values hold the source's time-of-flight delay back so it bends less in pitch, down to a barely audible bend at 0. Double-click to return to the default.

Progress

Under the options: the gesture being rendered right now, a bar with the share of it that is done, and the gestures still waiting behind it.

8.2 Stats

What the path adds up to, measured in the room's real metres around the listener. These update as you edit — they are a fast way to tell whether a gesture is a gentle drift or a violent flyby.

ROW	MEANING
Path length	Length of the path across the floor, in metres.
Ascent / descent	How far the source climbs and how far it descends along the path, in metres.
Avg / peak speed	Average and peak speed of the source along the path, in metres per second.
Accel / decel	Sharpest speed-up and slow-down, as multiples of the average speed.
Spatial centroid	Where the source spends its time on average, measured from the listener at the centre.

8.3 Source

The sound file this gesture plays: its name, the untrimmed waveform as it was imported, and a **trim button** in the corner that opens the trim editor. The trim button is unavailable until the gesture has a sound.

8.4 Mix

Five dials for the selected gesture. Double-click any of them to return it to 0 dB.

KNOB	RANGE	EFFECT
Gain	−24 ... +12 dB	Level of this gesture in the mix.
Low	−12 ... +12 dB	Cut or boost the low band.
Mid	−12 ... +12 dB	Cut or boost the mid band.
High	−12 ... +12 dB	Cut or boost the high band.
Send	−96 ... 0 dB	How much of this gesture is sent to the room reverb.

Mix changes are heard immediately — they are applied during playback and never trigger a re-render.

8.5 Master

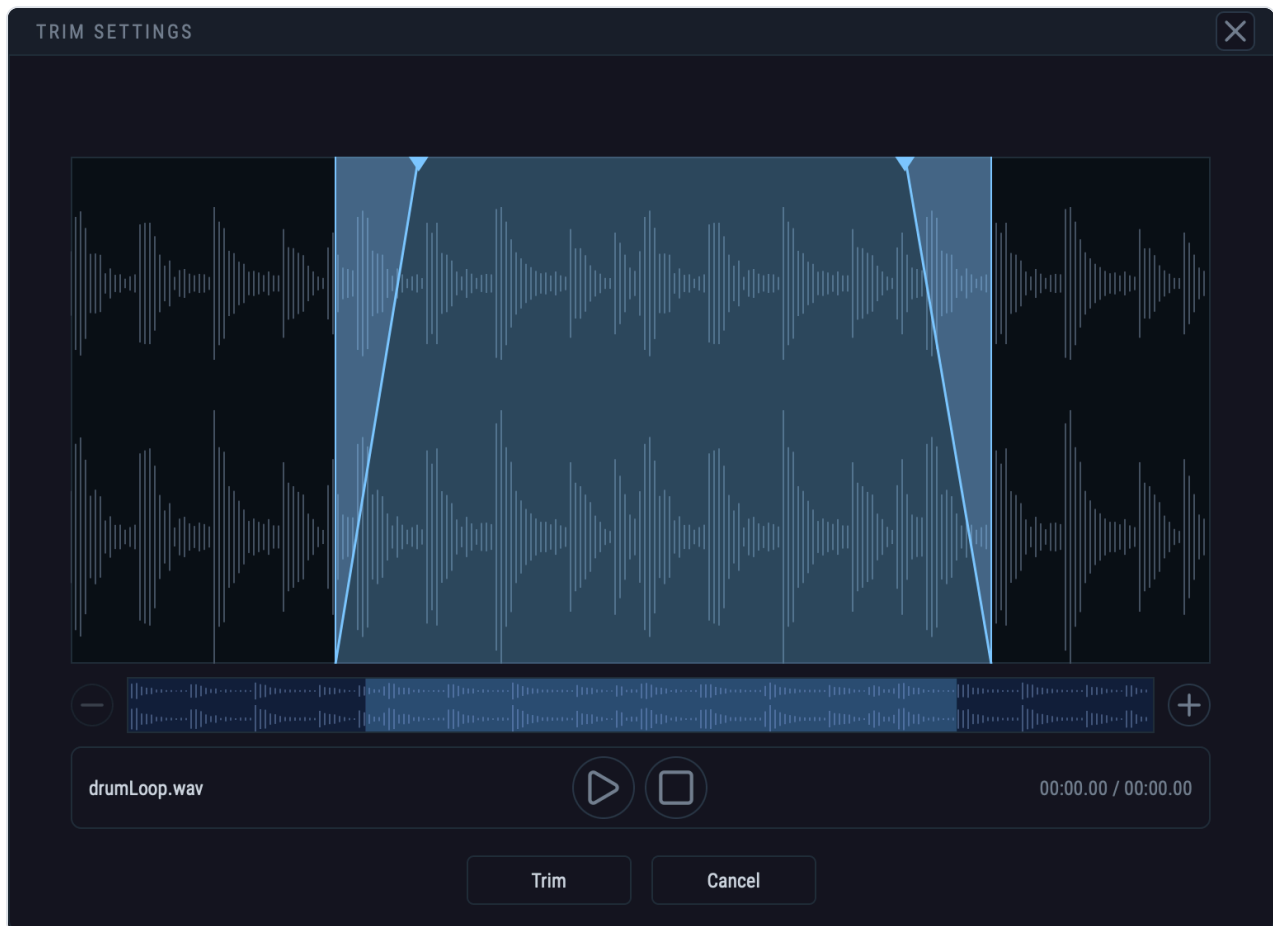
The room's output: what every gesture is mixed into.

- **Gain** — output level of the whole room, −24 dB to +12 dB. Double-click for 0 dB.

- **L / R meters** — peak level of each output channel.

9. Trimming a sound

The trim editor is an overlay opened from the trim button in the Source panel. It picks the part of the imported sound the gesture actually plays, and its fades.



The trim overlay. The lit region is what the gesture will play; the two diagonals sloping in from its edges are the fade in and fade out, dragged from the triangles at the top. The strip beneath is the whole file, with the kept part marked and the zoom buttons either side.

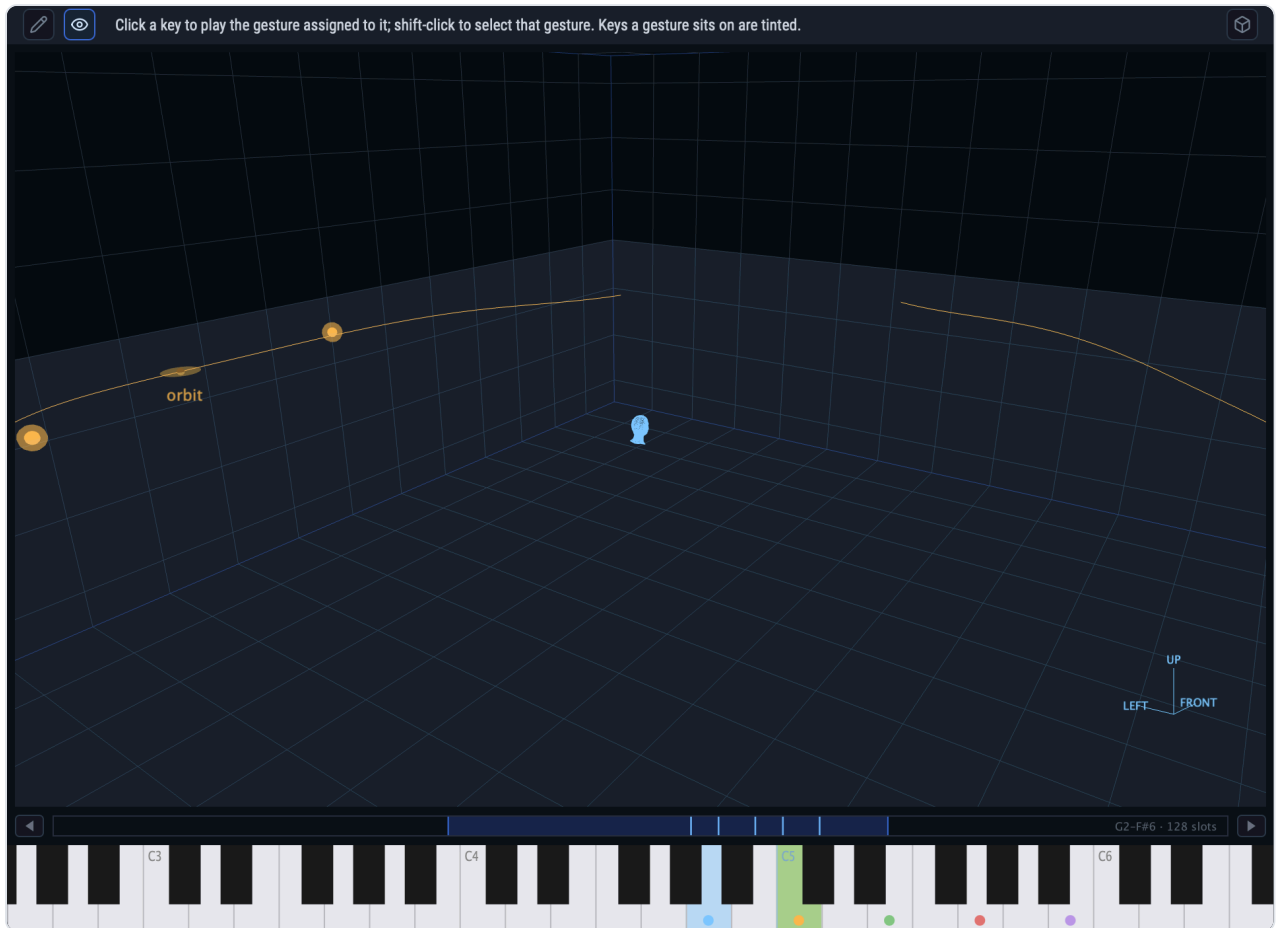
ELEMENT	WHAT IT DOES
Waveform	Drag to pick the part of the sound to keep. Drag its edges to resize the selection, its middle to slide it along, and the triangles to set the fade in and fade out.
Navigator strip	The whole sound file, with the part you are keeping marked along the bottom. Drag the lit band to move what is shown above; drag its edges to zoom.
– / + buttons	Show more of the sound, or less of it in more detail.
Play / stop	Hear the selected part with its fades applied, before committing.
Trim	Replaces the gesture's sound with the selected part. The gesture is re-rendered.
Cancel / x	Closes without changing the sound.

NON-DESTRUCTIVE

The imported file is never modified: SGD keeps the full import and writes the trimmed result alongside it. The Source panel always shows the untrimmed file, so you can come back and trim differently.

10. Observe view

The eye button in the header swaps the whole body for a 3D scene: the room seen from outside, with every *sounding* gesture drawn along its path in its own colour and a source marker walking it.



The observe view while orbit sounds. Its path is traced in the gesture's own colour and named at the node, the two markers are the left and right sources of a stereo gesture held apart by its width curve, and the ellipse beneath is the source's shadow on the floor. The listener stands at the centre; the gizmo bottom-right names the axes.

- **Drag** to orbit the camera around the room.
- **Scroll** to zoom in and out.
- An orientation gizmo in the corner shows which way the axes point.
- The box, its floor grid and the ear-level plane follow the room's real dimensions.

The observe view is a monitoring view, not an editing one: play keys (from the keyboard or your DAW) and watch. It is the quickest way to check that a gesture goes where you think it goes, and to see several gestures interact in the same space.

OVERLAYS AND 3D

The 3D views draw on a native surface that sits above ordinary widgets. This is why SGD prints tooltips in the header status line, and why the room preview steps aside when a confirmation question appears over it.

11. Playing & MIDI

A gesture is triggered by its key. There are two sources of keys, and they behave the same:

The on-screen keyboard

- **Click a key** to play the gesture assigned to it, for as long as the mouse is held.
- **Shift-click a key** to select that gesture in the gesture panel — a fast way to jump to a gesture you can hear but cannot find in a long list.
- Keys a gesture sits on are tinted in that gesture's colour. The selected gesture's key carries a wash of the accent colour; a key currently sounding carries a brighter one.

MIDI from the host

SGD accepts MIDI note on and note off from the DAW. A note on starts the gesture on that key; a note off stops it. Notes are placed **sample-accurately** — a note that arrives in the middle of a buffer starts at exactly that frame, and host frame positions are converted to engine frames on the way in, so timing is correct at any host sample rate.

Incoming DAW notes light up the on-screen keyboard exactly as clicks do, so you can see what the sequencer is playing.

NOTE ON THE NOTE

Velocity and pitch bend are not used: a gesture plays back at its own level, set by the Mix panel's Gain. Which note you send selects *which gesture*, not how loud or how high it is.

12. Audio, files & sessions

12.1 Sample rates

SGD's engine runs at a fixed **48 kHz** — the rate the binaural (HRTF) data exists at. Two conversions bridge that to the outside world:

- **On import.** A file at 44.1 kHz or 96 kHz is converted once, at import, to 48 kHz and stored as 32-bit float WAV. Everything held in memory and on disk is therefore 48 kHz.
- **On output.** The finished mix is converted from 48 kHz to the host's rate at the very last stage. At 48 kHz this is bypassed bit-exactly. The conversion's latency is reported to the host for delay compensation.

The practical upshot: your session rate does not change how anything inside SGD sounds, and a project moved between a 44.1 kHz and a 96 kHz session behaves identically.

12.2 Where things are stored

Rooms live in SGD's own application data folder, not next to your DAW project:

```
macOS    ~/Library/Application Support/Radical Process/SGD/
Windows  %APPDATA%\Radical Process\SGD\

<room-uuid>/
  room.json          the room: name, size, acoustics, every gesture
  samples/           imported sound files, converted to 48 kHz
  dry/               trimmed sounds, with fades applied
  binauralized/       the offline-rendered binaural result per gesture
```

A room folder is self-contained: it can be copied to another machine, and the room will open there with its sounds intact.

12.3 What the DAW saves

Your DAW session stores **only the identifier of the room that was open**. The room's own contents are saved into its `room.json` whenever the host asks the plugin for its state — that is, when you save the session — and whenever you press Save in the room editor.

CONSEQUENCE

A session file alone is not a backup. To archive a project, keep the matching room folder from the application data directory alongside it. Equally, two sessions that refer to the same room share it: a change made in one is visible in the other.

13. Reference

13.1 Parameter ranges

PARAMETER	WHERE	RANGE	DEFAULT
Gain	Mix	−24 ... +12 dB	0 dB
Low / Mid / High	Mix	−12 ... +12 dB	0 dB
Send	Mix	−96 ... 0 dB	0 dB
Master gain	Master	−24 ... +12 dB	0 dB
Doppler	Render	0 ... 100 %	100 %
Width / Height / Depth	Room	1 ... 30 m	12 m
Absorb	Room	0 ... 1	0.20
Damping	Room	0 ... 1	0.40
Scatter	Room	0 ... 1	0.50
RT60	Room	0.1 ... 10 s	1.5 s
Stereo width	Editor	−180 ... +180°	0°
Gesture key	Gesture card	0 ... 127 (C-1 ... G9)	from A4
Gesture name	Gesture card	≤ 20 characters	—

13.2 Room size presets

Thirteen presets set Width, Height and Depth to a typical space. Any manual change to a slider deselects the preset, which then reads *Custom*.

PRESET	WIDTH	HEIGHT	DEPTH
Booth	1.5 m	2.2 m	1.5 m
Car	1.8 m	1.2 m	3.0 m
Bathroom	2.0 m	2.4 m	2.5 m
Bedroom	3.0 m	2.4 m	3.2 m
Small room	3.5 m	2.4 m	4.0 m
Kitchen	4.0 m	2.5 m	3.5 m
Studio	6.0 m	3.0 m	5.0 m
Large room	10.0 m	3.5 m	8.0 m
Corridor	2.0 m	2.6 m	20.0 m
Live venue	20.0 m	8.0 m	25.0 m
Church	15.0 m	12.0 m	25.0 m
Cathedral	25.0 m	22.0 m	30.0 m
Station	30.0 m	15.0 m	28.0 m

13.3 Wall materials

Seven presets set Absorb, Damping, Scatter and RT60 to a rough reference for a surface, ordered here from most reflective to most absorbent:

MATERIAL	ABSORB	DAMPING	SCATTER	RT60
Glass	0.05	0.05	0.05	3.0 s
Brick	0.04	0.15	0.40	3.5 s
Concrete	0.05	0.10	0.10	4.0 s
Wood	0.15	0.30	0.30	1.8 s
Carpet	0.40	0.70	0.20	0.8 s
Curtain	0.50	0.80	0.30	0.6 s
Acoustic foam	0.85	0.90	0.25	0.3 s

Concrete rings longest and scatters least; brick is the same hardness with rougher walls, so it diffuses more. Wood is the middle ground. Carpet, curtain and foam swallow progressively more and take the highs first, so the tail shortens and darkens together.

13.4 Mouse & keyboard

WHERE	ACTION	RESULT
Top view	Click empty space	Append a trajectory node
Top view	Drag node / handle	Move the node / curve the segment
Top view	Shift -click node	Remove the node
Elevation	Drag node up / down	Raise or lower the path there
Elevation	Drag node sideways	Slide the node across the room
Elevation	Shift -click node	Remove the node
Motion / Width	Click	Add a node
Motion / Width	Drag	Move a node
Motion / Width	Shift -click	Remove an interior node
Any knob	Double-click	Return to its default value
Gesture card	Click	Select the gesture
Gesture card	Click the colour stripe	Open the colour palette
Gesture card	Click the name	Rename the gesture
Gesture card	Drop a file	Import it as this gesture's sound
Keyboard	Click a key	Play the gesture on that key
Keyboard	Shift -click a key	Select that gesture
3D views	Drag	Orbit the camera
3D views	Scroll	Zoom in / out
Trim waveform	Drag	Select the part to keep
Trim waveform	Drag the triangles	Set fade in / fade out
Anywhere	Hover	Read the description in the header status line

14. Troubleshooting

The plugin opened with a room dialog I cannot close

That is by design: SGD needs a room before it can do anything, and no room exists yet. Name it, press Save, and Cancel becomes available from then on.

I pressed a key and heard nothing

Check, in order: the gesture has a sound (the card shows a file name, not a Load button); the card is not still showing QUEUED or a spinner; its Mix Gain is not at the bottom of its range; the Master gain is up and the meters move. Also confirm you are sending the note the card's key chip names.

My edit did not change what I hear

Edits to the path, motion, width, trim or render options need a re-render. Watch the status badge on the card and the Render panel; when it reads READY, play again. Mix, master and the room's acoustics, by contrast, are heard immediately.

The spatial effect is weak or confusing

Listen on headphones — the render is binaural. Then check the *Direct* toggle is on, that the path actually moves away from the centre (see Path length in Stats), and that the room is big enough for the movement to mean anything.

A moving sound warbles or sweeps in pitch

That is the Doppler shift of a fast source. Flatten the motion curve, lengthen the path, or turn the Render panel's **Doppler** knob down.

My file was refused

SGD accepts a single mono or stereo .wav / .aiff file. Multi-channel files, other formats, and multi-file drops are rejected.

Rendering is slow

Render time grows with the length of the sound and with each reflection order enabled. Turn *3rd* off while designing, and switch it on for the final pass.

I opened an old session and the room is empty or missing

The session stores only the room's identifier; the room itself lives in the application data folder. If that folder was cleared, or the session was opened on another machine, copy the room folder across (see Where things are stored).

SGD is slightly late against the rest of the session

SGD reports its output-conversion latency to the host. Make sure delay compensation is enabled in your DAW. Running the session at 48 kHz removes the conversion — and the latency — entirely.

15. Glossary

TERM	MEANING
Binaural	Rendered for two ears over headphones, using head-related transfer functions, so sounds appear at real positions around the listener.
Dry	The trimmed sound without spatialisation. It is what feeds the reverb send.
Early reflections	The first few bounces off the walls, rendered per gesture and selected by the Render panel's order toggles.
Gesture	A sound plus the path it travels, its timing, its mix and its key.
Key / slot	The MIDI note a gesture is bound to. 128 exist per room; no two gestures share one.
Motion curve	The map from playback time to position along the path.
Room	The space, its acoustics, and everything inside it. One is open at a time.
RT60	The time the reverb tail takes to fall by 60 dB.
Trajectory	The curved path through the room, with a height at every node.
Trim	The excerpt of the imported file the gesture plays, with fades.

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Behaviour described here matches the shipped build; hover any control to read its own description in the header status line.